

A Bicomplete Baire Quasi-Metric with Zero Symmetry Index

Run `fa.banach_001`, lane 5

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Status

Candidate counterexample, likely valid. The construction below gives negative answers to both Open Questions 1 and 2 of Bachir's *Asymmetric Normed Baire Space* [1].

Source Problems

The source defines

$$c(X) = \inf_{d(y,x) > 0} \frac{d(x,y)}{d(y,x)}$$

and asks on PDF page 6 whether every quasi-hemi-metric space with $c(X) = 0$ must fail to be Baire:

the above theorem, $(l^\infty(\mathbb{N}^*), \|\cdot\|_\infty)$ is not a Baire space.

Remark 1. The above theorem extends easily to asymmetric normed cone with the same proof, replacing "biBanach" by "bicomplete" (we refer to [7, 5] for more informations about asymmetric normed cones and their duals).

Notice that it is easy to see that for any quasi-hemi-metric space (X, d) , we have that $c(X)d_s \leq d \leq d_s$ ($c(X)$ may be equal to zero). Thus, if $c(X) > 0$ and (X, d) is bicomplete (equivalently (X, d_s) is a complete metric space), then (X, d) is a Baire space as d is equivalent to the complete metric d_s . We don't know if the converse remains true. We have the following open question.

Open question 1. Does the result of Theorem 2 extend to any quasi-hemi-metric space? More precisely, if (X, d) is any quasi-hemi-metric space, does $c(X) = 0$ imply that (X, d) is not a Baire space?

We are going to prove in Theorem 3 below, that the condition $c(X) = 0$ in quasi-hemi-metric space, is however always equivalent to the fact that the semi-Lipschitz free space $\mathcal{F}_a(X)$ over (X, d) (introduced recently in [7]) is not a Baire space. We recall below the notion of semi-Lipschitz free space and we refer to [7] for more details.

The semi-Lipschitz free space. The semi-Lipschitz free space over a quasi-hemi-metric space $X_d := (X, d)$ was recently introduced and studied in [7]. Its construction is analogous to the classical Lipschitz free space over a metric

On PDF page 8 it asks whether the semi-Lipschitz free space $\mathcal{F}_a(X)$ of a Baire quasi-hemi-metric space must itself be Baire. The source's Theorem 3, shown in the same crop, states that $c(X) = 0$ if and only if $\mathcal{F}_a(X)$ is not Baire.

Theorem 3. *Let (X, d) be a quasi-hemi-metric space and $\mathcal{F}_a(X)$ be the semi-Lipschitz free space over (X, d) . Then, $c(X) = 0$ if and only if $(\mathcal{F}_a(X), \|\cdot\|_*)$ is not a Baire space.*

Proof. The proof follows from Proposition 2 and Theorem 2, noting that $(\mathcal{F}_a(X), \|\cdot\|_*)$ is a biBanach space by construction. $\square$

From the above result, we can reformulate the remark just before the Open question 1, as follows.

Proposition 3. *Suppose that (X, d) is a bicomplete quasi-hemi-metric space such that $(\mathcal{F}_a(X), \|\cdot\|_*)$ is a Baire space. Then (X, d) is also a Baire space.*

We don't know if the converse in the above proposition is true in general.

Open question 2. Let (X, d) be a quasi-hemi-metric space. Suppose that (X, d) is a Baire space, does $(\mathcal{F}_a(X), \|\cdot\|_*)$ is also a Baire space?

It is clear that Open question 1 and Open question 2, are in fact equivalent.

DECLARATION

The author declare that there is no conflict of interest.

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Definitions

For $a \in \mathbb{R}$, write $a_+ = \max\{a, 0\}$. A quasi-hemi-metric satisfies the triangle inequality and separates distinct points after considering both directions. A quasi-metric satisfies the stronger implication $d(x, y) = 0$ only when $x = y$. Its forward balls are

$$B_d(x, r) = \{y : d(x, y) < r\}.$$

The symmetrized metric is $d_s(x, y) = \max\{d(x, y), d(y, x)\}$, and the space is bicomplete when d_s is complete.

Idea of the Proof

A directed potential $q_f(x, y) = (f(y) - f(x))_+$ always satisfies the triangle inequality. We add the potentials $f(t) = t$ and $g(t) = e^{-t}$. The first charges motion to the right, while the second charges motion to the left. Both charges are locally comparable to Euclidean distance, so the forward topology remains Euclidean. Far to the right, however, the leftward charge across a unit interval is exponentially small while the rightward charge is one. This destroys every global lower symmetry bound and forces $c(X) = 0$.

Counterexample

Theorem 1. *Define $d : \mathbb{R} \times \mathbb{R} \rightarrow [0, \infty)$ by*

$$d(x, y) = (y - x)_+ + (e^{-y} - e^{-x})_+.$$

Then:

1. d is a quasi-metric;
2. the topology generated by the forward d -balls is the usual Euclidean topology on $\mathbb{R}$, and hence $(\mathbb{R}, d)$ is a Baire space;
3. $c(\mathbb{R}, d) = 0$;
4. the symmetrized metric is

$$d_s(x, y) = \max\{|x - y|, |e^{-x} - e^{-y}|\},$$

and it is complete.

Consequently, the implication proposed in Open Question 1 is false, even among bicomplete quasi-metric spaces. Moreover, $\mathcal{F}_a(\mathbb{R}, d)$ is not Baire, so Open Question 2 is also false.

Proof. For an arbitrary real-valued function f on a set, put

$$q_f(x, y) = (f(y) - f(x))_+.$$

The elementary inequality $(u + v)_+ \leq u_+ + v_+$ gives

$$q_f(x, y) \leq q_f(x, z) + q_f(z, y).$$

Thus every q_f satisfies the triangle inequality. Our function is the sum of q_f for $f(t) = t$ and $f(t) = e^{-t}$, so d satisfies the triangle inequality. Also $d(x, x) = 0$. If $x < y$, then the first summand is $y - x > 0$; if $x > y$, then the second summand is $e^{-y} - e^{-x} > 0$. Hence $d(x, y) = 0$ implies $x = y$, and d is a quasi-metric.

For fixed x , the definition simplifies to

$$d(x, y) = \begin{cases} y - x, & y \geq x, \\ e^{-y} - e^{-x}, & y < x. \end{cases} \quad (1)$$

The map $y \mapsto d(x, y)$ is Euclidean-continuous, so each forward ball $B_d(x, r)$ is Euclidean-open. It follows that the d -topology is no finer than the Euclidean topology.

Conversely, fix $x \in \mathbb{R}$ and $\varepsilon > 0$, and set

$$r = \min \left\{ \varepsilon, e^{-(x-\varepsilon)} - e^{-x} \right\} > 0.$$

If $y \geq x$ and $d(x, y) < r$, then by (1), $0 \leq y - x < r \leq \varepsilon$. If $y < x$ but $y \leq x - \varepsilon$, then

$$d(x, y) = e^{-y} - e^{-x} \geq e^{-(x-\varepsilon)} - e^{-x} \geq r,$$

a contradiction. Therefore

$$B_d(x, r) \subset (x - \varepsilon, x + \varepsilon).$$

Every Euclidean neighborhood of x thus contains a forward d -ball, so the two topologies coincide. The real line with its Euclidean topology is a Baire space; hence $(\mathbb{R}, d)$ is Baire.

Now take $x_n = n + 1$ and $y_n = n$, for $n \geq 0$. Formula (1) gives

$$d(y_n, x_n) = 1, \quad d(x_n, y_n) = e^{-n} - e^{-(n+1)} = (1 - e^{-1})e^{-n}.$$

The denominators $d(y_n, x_n)$ are positive, and therefore

$$0 \leq c(\mathbb{R}, d) \leq \frac{d(x_n, y_n)}{d(y_n, x_n)} = (1 - e^{-1})e^{-n} \rightarrow 0.$$

Thus $c(\mathbb{R}, d) = 0$.

If $x < y$, then $d(x, y) = y - x$ and $d(y, x) = e^{-x} - e^{-y}$. Interchanging x, y proves for all pairs that

$$d_s(x, y) = \max\{|x - y|, |e^{-x} - e^{-y}|\}.$$

If (x_k) is d_s -Cauchy, then it is Euclidean-Cauchy because $|x_k - x_\ell| \leq d_s(x_k, x_\ell)$. Hence $x_k \rightarrow x \in \mathbb{R}$. Continuity of the exponential then gives

$$d_s(x_k, x) = \max\{|x_k - x|, |e^{-x_k} - e^{-x}|\} \rightarrow 0.$$

Thus d_s is complete and the example is bicomplete.

We have now produced a Baire quasi-metric space with $c(X) = 0$, disproving Open Question 1. Finally, Theorem 3 of the source paper [1] states that $c(X) = 0$ is equivalent to the asymmetric normed space $\mathcal{F}_a(X)$ not being Baire. Applying it here shows that $\mathcal{F}_a(\mathbb{R}, d)$ is not Baire even though $(\mathbb{R}, d)$ is Baire. This disproves Open Question 2. $\square$

Remark 1. *The example is deliberately outside the asymmetric normed setting of the source’s Theorem 2. The function d is not translation invariant: the cost of moving one unit to the left is $(1 - e^{-1})e^{-x}$, which depends on the base point x . Thus the counterexample does not contradict that theorem; it shows exactly why its linear homogeneity cannot be dropped.*

Verification Notes

The proof is elementary and self-contained apart from the source’s Theorem 3, which is used only to transfer the counterexample from Open Question 1 to Open Question 2. The script `code/check_counterexample.py` checks the triangle inequality on a 13-point grid and 100,000 seeded random triples, checks the displayed formula for d_s , and prints the symmetry ratios for unit intervals. It completed with no violation. These finite tests are sanity checks, not a substitute for the universal argument above.

Novelty and Limitations

A bounded search on August 9, 2026 found no duplicate in the run indexes for arXiv:2105.04862, quasi-hemi-metric Baire spaces, the symmetry index, or semi-Lipschitz free spaces. Web searches used the exact title and DOI, the phrases “Open question 1”, “ $c(X) = 0$ Baire quasi-metric”, “index of symmetry quasi-metric Baire”, and “semi-Lipschitz free space Baire”. They returned the source, bibliographic mirrors, and later background citations, but no later claimed resolution. This is a bounded novelty check, not an exhaustive citation or MathSciNet review.

The conclusion for Open Question 2 relies on Theorem 3 of the source paper. The construction itself directly and independently settles Open Question 1.

Human Review Recommendation

Accept as a likely valid counterexample after checking the orientation in the definition of $c(X)$, the two topology inclusions, and the invocation of source Theorem 3. There are no conditional lemmas or numerical dependencies.

References

- [1] M. Bachir, *Asymmetric Normed Baire Space*, Results in Mathematics **76** (2021), article 176; arXiv:2105.04862v2, <https://doi.org/10.1007/s00025-021-01483-6>.